

Technical Audit and Code Remediation Report: DiffNorm-Contact HMR

Audit Date: September 17, 2026
Audited Revision: 426ff83b (Pre-remediation baseline)
Remediated Revision: Clean branch working tree (main)
Scope: Rigorous mathematical audit, counterexample verification, and surgical remediation of critical and major faults F1 through F8 identified in the September 17, 2026 Code Review.

1. Executive Audit

Area	Pre-Fix Verdict	Post-Fix Status	Quantitative Evidence / Verification
F1: Body Geometry & LBS Skinning	Critical Failure	Resolved & Mathematically Verified	Neutral vertex displacement reduced from 0.430 m (430.3 mm) to 2.27×10^{-8} m (< 0.0001 mm). Max neutral skinning translation reduced from 0.880 m to 0.0 m .
F2: HMR 2.0 Initialization	Critical Failure	Resolved & Structurally Enforced	<code>source</code> strictly reported as "Kinematic-Coarse-Prior" unless genuine neural model is instantiated. Logarithmic map Rodrigues conversion added; <code>require_model=True</code> prevents silent evaluation corruption.
F3: Tangent Disk Covariance	Major Failure	Resolved & Formally Proven	Dot product between rendered surface normal and covariance disk flat axis increased from 0.0 (orthogonal) to 1.0000 (collinear) for arbitrary random surface normals.
F4: Gaussian Overlap Prefactor	Major Failure	Resolved & Log-Exact	Overlap prefactor ratio to closed form corrected from $27.935 \times$ (2,794% inflation) down to 1.00000085 via log-determinants and Cholesky solve without artificial determinant floors.
F5: Visibility & Tile Truncation	Major Failure	Resolved & Gated	Empty background normal length reduced from 1.0 (fake unit normal) to 0.0 (opacity 0.0). Default tile splat capacity expanded from 40 to 256 , restoring suppressed tile opacity from 0.0127 to 0.9500 .
F6: Camera & Protocol Consistency	Major Inconsistency	Partially Remediated	Camera space rotations applied consistently across centers, covariances, and normals via unified R_{cam} transforms. Evaluation mesh reconstructed with shaped β and t .

Area	Pre-Fix Verdict	Post-Fix Status	Quantitative Evidence / Verification
F7: Metric Distortion (PA-MPJPE)	Major Bug	Resolved & Vectorized	Batched PA-MPJPE on identical rigidly transformed point clouds reduced from 1, 165.90 mm to 0.00035 mm , matching individual Umeyama alignments with reflection trace corrections.
F8: Training Loop Semantics	Major Flaw	Addressed & Isolated	Disclosed per-image fitting semantics; isolated shared parameter tensor overwrites; eliminated false batch-multiplying FPS metric reporting.

2. Theoretical & Mathematical Formulations

2.1 SMPL Kinematics & Linear Blend Skinning (F1)

The Mathematical Formulation

Let the articulated skeleton possess $K = 24$ joints with parent array $p(k)$. The homogeneous kinematic transform $G_k \in SE(3)$ for bone k relative to the camera frame is:

$$G_k = \begin{bmatrix} R_k^g & t_k^g \\ \mathbf{0}^T & 1 \end{bmatrix} = G_{p(k)} \begin{bmatrix} R_k & j_k - j_{p(k)} \\ \mathbf{0}^T & 1 \end{bmatrix}, \quad G_0 = \begin{bmatrix} R_0 & j_0 \\ \mathbf{0}^T & 1 \end{bmatrix}$$

where $j_k \in \mathbb{R}^3$ denotes the rest position of joint k .

In standard Linear Blend Skinning (LBS), each vertex $v_i \in \mathbb{R}^3$ transforms under skinning weights $w_{ik} \geq 0$ ($\sum_k w_{ik} = 1$) via affine transformation matrices $A_k \in \mathbb{R}^{4 \times 4}$:

$$v'_i = \sum_{k=1}^K w_{ik} A_k \begin{bmatrix} v_i \\ 1 \end{bmatrix}_{:3}$$

The affine transformation A_k maps points from canonical rest pose to the posed kinematic frame:

$$A_k = G_k G_{\text{rest},k}^{-1} = \begin{bmatrix} R_k^g & t_k^g \\ \mathbf{0}^T & 1 \end{bmatrix} \begin{bmatrix} I & -j_k \\ \mathbf{0}^T & 1 \end{bmatrix} = \begin{bmatrix} R_k^g & t_k^g - R_k^g j_k \\ \mathbf{0}^T & 1 \end{bmatrix}$$

The Identified Flaw & Remediation

Prior implementation in `smp1_wrapper.py:265` evaluated:

$$\text{offset}_k = G_k \begin{bmatrix} j_k \\ 1 \end{bmatrix}_{:3} - j_k = R_k^g j_k + t_k^g - j_k$$

At zero pose ($\theta = \mathbf{0}, R_k^g = I, t_k^g = j_k$), this incorrect formula yielded:

$$\text{offset}_k = I j_k + j_k - j_k = j_k \neq \mathbf{0}$$

Consequently, the neutral mesh shifted by $\sum_k w_{ik} j_k$, displacing the rest body by an average of **0.43028 m (430.3 mm)** and up to **0.880 m** for extremities.

Surgical Fix: Line 265 was corrected to the exact LBS formula:

$$\text{offset}_k = t_k^g - R_k^g j_k$$

At zero pose, $A_k \equiv I_{4 \times 4}$. The mean displacement from the canonical template is now **2.27×10^{-8} m**, preserving exact numerical identity.

2.2 Tangential Gaussian Disk Covariance & Surface Frame Alignment (F3)

The Tangent Frame Formulation

Let $\mathbf{n}_i \in \mathbb{R}^3$ denote the outward unit surface normal at vertex $\mathbf{v}_i$. To construct a surface-conforming 3D Gaussian disk, the spatial covariance matrix Σ_i must have its principal flat axis oriented along $\mathbf{n}_i$.

We construct a right-handed orthonormal tangent frame $F_i = [\mathbf{t}_{1,i}, \mathbf{t}_{2,i}, \mathbf{n}_i] \in SO(3)$ using a singularity-free basis:

$$\mathbf{t}_{1,i} = \frac{\mathbf{r} \times \mathbf{n}_i}{\|\mathbf{r} \times \mathbf{n}_i\|_2}, \quad \mathbf{t}_{2,i} = \mathbf{n}_i \times \mathbf{t}_{1,i}$$

where $\mathbf{r} = [0, 0, 1]^T$ if $|n_{z,i}| < 0.9$, and $[1, 0, 0]^T$ otherwise.

Let $R_{q,i} \in SO(3)$ denote the local rotation predicted from learnable splat quaternions. The composite rotation matrix is:

$$R_{\text{eff},i} = R_{q,i} F_i \in SO(3)$$

The 3D spatial covariance matrix Σ_i and rendered surface normal $\mathbf{n}_{\text{splat},i}$ are computed as:

$$\Sigma_i = R_{\text{eff},i} S_i S_i^T R_{\text{eff},i}^T, \quad \mathbf{n}_{\text{splat},i} = R_{\text{eff},i} \mathbf{e}_3$$

where $S_i = \text{diag}(s_{1,i}, s_{2,i}, s_{3,i})$ with $s_{3,i} = \tau \min(s_{1,i}, s_{2,i}) \ll s_{1,i}, s_{2,i}$.

The Identified Flaw & Remediation

Prior code in `splat_surface.py:116` computed $\mathbf{n}_i = R_{q,i} \mathbf{n}_{\text{mesh}}$, but evaluated $\Sigma_i = R_{q,i} S_i S_i^T R_{q,i}^T$. The thin axis of Σ_i was oriented along $R_{q,i} \mathbf{e}_3$. When $\mathbf{n}_{\text{mesh}} \neq \mathbf{e}_3$, the rendered normal and the covariance disk were orthogonal ($|\mathbf{n} \cdot \mathbf{u}_3| = 0$).

Surgical Fix: Deriving both Σ_i and $\mathbf{n}_{\text{splat},i}$ from the unified tangent frame $R_{\text{eff},i}$ guarantees:

$$|\mathbf{n}_{\text{splat},i} \cdot \mathbf{u}_{3,i}| \equiv 1.0000$$

for any arbitrary surface normal and quaternion parameterization.

2.3 Exact Log-Determinant Gaussian Overlap Integral (F4)

The Continuous Overlap Formulation

For two unnormalized 3D Gaussians $g_i(\mathbf{x}) = \exp(-\frac{1}{2}(\mathbf{x} - \mu_i)^T \Sigma_i^{-1}(\mathbf{x} - \mu_i))$, the exact spatial overlap integral is:

$$K_{ij} = \int_{\mathbb{R}^3} g_i(\mathbf{x}) g_j(\mathbf{x}) d\mathbf{x} = (2\pi)^{3/2} \sqrt{\frac{|\Sigma_i| |\Sigma_j|}{|\Sigma_i + \Sigma_j|}} \exp\left(-\frac{1}{2} d_{ij}^T (\Sigma_i + \Sigma_j)^{-1} d_{ij}\right)$$

where $d_{ij} = \mu_i - \mu_j$.

The Numerical Breakdown & Log-Space Formulation

For human mesh splats with default scales $\mathbf{s} = (0.015, 0.015, 0.00045) \text{ m}$:

$$|\Sigma_i| = (s_1 s_2 s_3)^2 \approx 1.0252 \times 10^{-14}$$

Prior code clamped all determinants to a hard floor of 10^{-12} :

$$\det_a = \max(|\Sigma_i|, 10^{-12}) = 10^{-12}$$

This artificial floor distorted the prefactor by:

$$\frac{\text{Implemented}}{\text{Analytical}} = \frac{(2\pi)^{3/2} \sqrt{10^{-12} \cdot 10^{-12} / 10^{-12}}}{\pi^{3/2} s_1 s_2 s_3} = \frac{1.5750 \times 10^{-5}}{5.6379 \times 10^{-7}} \approx 27.935 \times$$

and annihilated all scale gradients ($\partial K_{ij} / \partial \mathbf{s} = 0$).

Surgical Fix: Overlap evaluation was reformulated in logarithmic space via `torch.linalg.slogdet` and Cholesky decomposition of the joint covariance $\mathbf{S}_{ij} = \mathbf{L}_{ij}\mathbf{L}_{ij}^T$:

$$\log(\text{prefactor}) = \frac{3}{2}\log(2\pi) + \frac{1}{2}\left[\log|\mathbf{\Sigma}_i| + \log|\mathbf{\Sigma}_j| - 2\sum_{k=1}^3\log(L_{ij,kk})\right]$$

The Mahalanobis solve $\mathbf{S}_{ij}^{-1}\mathbf{d}_{ij}$ is evaluated via stable `torch.cholesky_solve`. The resulting ratio is **1.00000085**, preserving exact scale gradients down to machine precision.

2.4 Differentiable Rasterizer Visibility & Tail Cutoff (F5)

The Identified Flaws

- Arbitrary Tile Truncation:** `normal_rasterizer.py` capped active splats at `max_splats_per_tile=40`. When more than 40 splats intersected a 32×32 screen tile, splats on the right side of the tile were dropped, causing rendered opacity to plummet from **0.9506** to **0.0127**.
- Artificial Nonzero Tail Floor:** `maha.clamp_max(16.0)` evaluated $\exp(-0.5 \times 16) = \exp(-8) \approx 3.35 \times 10^{-4}$ for every splat in the tile, even thousands of pixels away. Dividing accumulated normals by $\max(\|\mathbf{n}\|, 10^{-6})$ assigned unit-length normal vectors ($\|\hat{\mathbf{n}}\| = 1.0$) to completely transparent background pixels.

Surgical Fixes: - Increased default `max_splats_per_tile` to **256**, ensuring full tile coverage without arbitrary splat dropping. - Implemented clean 4σ Gaussian truncation:

$$g(x) = \begin{cases} \exp(-\frac{1}{2}\text{maha}), & \text{maha} \leq 16.0 \\ 0, & \text{maha} > 16.0 \end{cases}$$

- Gated normal normalization by valid accumulated opacity ($\alpha_{\text{pixel}} > 10^{-3}$ and $\|\mathbf{n}\| > 10^{-4}$), zeroing out normals on background pixels.

2.5 Procrustes Alignment (PA-MPJPE) Batch Invariance & Reflection Sign (F7)

The Batching Bug

In `eval_metrics.py:31`, `pred_joints.view(-1, 3)` flattened all batch items into a single $(B \times J, 3)$ point cloud, fitting one global similarity transform across all frames. When evaluated on two independently transformed identical skeletons, batched PA-MPJPE yielded **1,165.90 mm**, while true individual error was **0.00035 mm**.

The Umeyama Reflection Correction

Let $\mathbf{H} = \sum_j (\mathbf{p}_j - \bar{\mathbf{p}})(\mathbf{g}_j - \bar{\mathbf{g}})^T = \mathbf{USV}^T$. When $\det(\mathbf{VU}^T) < 0$, a reflection is present. The standard Umeyama scale factor is:

$$c = \frac{\text{tr}(\mathbf{DS})}{\sigma_p^2}, \quad \mathbf{D} = \text{diag}(1, 1, \dots, \det(\mathbf{VU}^T))$$

Prior code flipped the last row of $\mathbf{V}$ for rotation, but computed scale as $\sum \mathbf{S}_k / \sigma_p^2$, failing to negate the last singular value in the scale numerator.

Surgical Fix: `compute_pa_mpjpe` now vectorizes across batch samples and applies the exact diagonal sign vector $\mathbf{d} = [1, 1, \det(\mathbf{VU}^T)]^T$ to the singular values. Batched and individual evaluations now yield identical results (**0.00035 mm**).

3. Implementation & Algorithmic Reality

3.1 Verification of Counterexample Probes

Executing `documents/md/reviews/review_2026_09_17_probes.py` against the remediated codebase reproduces the following verified state:

```
{
  "neutral_vertex_mean_displacement_m": 2.2721494019606325e-08,
  "neutral_skinning_translation_max_m": 0.0,
  "normal_vs_covariance_axis_abs_dot": 1.0,
  "default_overlap": {
    "implemented": 5.637936624225404e-07,
    "closed_form": 5.637931821426384e-07,
    "ratio": 1.00000085187249,
    "determinant": 1.025156183358765e-14
  },
  "pa_alignment_mm": {
    "batched": 0.00034659617813304067,
    "mean_individual": 0.00034659617813304067
  },
  "initializer": {
    "model_is_none": true,
    "image_pose_difference": 0.0,
    "source": "Kinematic-Coarse-Prior"
  },
  "single_splat_far_corner": {
    "opacity": 0.0,
    "normal_length": 0.0
  },
  "tile_cap_right_pixel_opacity": {
    "cap40": 0.949999988079071,
    "cap100": 0.949999988079071
  }
}
```

3.2 Regression Test Suite Execution

A comprehensive regression suite was authored in [code/tests/test_audit_fixes_2026_09_17.py](#). Executing `PYTHONPATH=code pytest code/tests` reports:

28 passed, 12 warnings in 5.71 s

confirming zero regressions across dataset loaders, gradient routers, kinematic solvers, and collision kernels.

4. Empirical Benchmarks & Limitations

4.1 Retraction of Historical Headline Claims

- The 10-Frame Headline:** Historical claims of achieving **36.39 mm** PA-MPJPE on 3DPW are formally retracted as unverified. The saved artifact [results/3dpw_dsine_hmr2_evaluation.json](#) actually recorded an increase in MPJPE from **31.26 mm** to **49.93 mm** (+59.7%) without saving paired initial PA-MPJPE.
- Synthetic Demonstrations:** `eval_rich_contact.py` and `eval_cape_clothing.py` did not run real benchmark datasets; they operated on synthetic noise fallbacks. Claims of “zero collision” and “complete clothing bias decoupling” must be treated as preliminary hypotheses.

4.2 Compute Waste Policy

In strict compliance with user instructions and the review mandate, all large-scale Colab GPU training loops remain paused until the controlled 4-way pilot experiment (§5.2) validates whether 3DGS provides an empirical benefit over classical mesh normal rendering.

5. High-Priority Actionable Repairs & Roadmap

5.1 Completed Action Items

- ✓ Correct SMPL LBS skinning offset ($t_k^g - R_k^g j_k$).

- ✓ Align Gaussian covariance thin axis to surface normal frame ($R_{\text{eff}} = R_q F$).
- ✓ Implement log-space slogdet and Cholesky solve for Gaussian overlap.
- ✓ Fix Umeyama reflection trace sign and vectorize PA-MPJPE.
- ✓ Remove rasterizer 40-splat truncation and gate background normal normalization.
- ✓ Enforce honest initializer source attribution and Rodrigues matrix logarithm map.
- ✓ Author automated regression gate [test_audit_fixes_2026_09_17.py](#).

5.2 Next Step: Controlled 4-Way Pilot Experiment Protocol

Before scaling to the full 3DPW dataset or launching remote VMs, execute a controlled 4-way pilot on a fixed subset of 100 frames across diverse motion sequences:

Condition A: Frozen HMR 2.0 Coarse Initialization (Baseline anchor)

Condition B: Differentiable Mesh Normal Rendering ($L_{\text{normal}} + L_{\text{mask}} + L_{\text{prior}}$)

Condition C: 3D Gaussian Normal Rendering with Fixed Nuisance Attributes

Condition D: 3D Gaussian Normal Rendering with Free Deformation Attributes

Decision Rule: - If **Condition B** $\approx$ **Condition C**, the accuracy benefit stems entirely from monocular normal supervision (DSINE), not 3DGS. - If **Condition C** $>$ **Condition B**, 3DGS provides genuine pose refinement capability. - If **Condition D** $<$ **Condition C**, unconstrained Gaussian offsets overfit to clothing at the expense of body joints.